STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 1: Units and Measurements

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. How many SI base quantities? [1]
2. SI unit of temperature? [1]
3. Significant figures in 0.00450? [1]
4. Dimensions of speed? [1]
5. Dimensions of force? [1]
6. Is length/length dimensionless? [1]
7. Can dimensional analysis determine pi? [1]
8. SI unit of pressure? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. Round 12.486 to 3 significant figures. [2]
10. Dimensions of momentum mv? [2]
11. Check s=ut+at^2 dimensionally. [2]
12. Convert 72 km/h to m/s. [2]
13. Significant figures in 5.070 x 10^3? [2]
14. Why can’t dimensions distinguish work and torque? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. Derive pendulum period dependence T proportional to l^ag^b. [3]
16. Sphere radius 2.00 cm: volume to 3 s.f.? [3]
17. Dimensions of G from F=Gm_1m_2/r^2? [3]
18. 15.2 cm x 3.4 cm: area with correct precision? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Spring-mass period T proportional to m^ak^b, [k]=MT^-2. Find exponents. [4]
20. Calculated volume is 18.237 cm^3, least precise input 3 s.f. What is reported? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations worth knowing

- Speed: v = distance/time; velocity: v = displacement/time; acceleration: a = Delta v/Delta t.
- Dimensions: [v]=LT^-1; [a]=LT^-2; [F]=MLT^-2; [momentum]=MLT^-1; [pressure]=ML^-1T^-2; [G]=M^-1L^3T^-2.
- Dimensional check: every term in an equation must have identical dimensions.
- Dimensional method for T proportional to l^a g^b: [T]=L^a(LT^-2)^b, so a+b=0 and -2b=1.
- Spring period dependence: T proportional to m^a k^b with [k]=MT^-2; compare mass and time powers.
- Sphere volume V = (4/3) pi r^3; rectangle area A = length x breadth.
- Unit conversion: 1 m/s = 3.6 km/h; 1 km/h = 5/18 m/s.
- Rounding: keep the requested significant figures; for multiplication/division, report to the least precise input’s significant figures.

Use each equation only when its assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Solutions are grouped here after the question paper and formula reference.

Question 1

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Seven.
Final answer: Seven.

Question 2

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: kelvin (K).
Final answer: kelvin (K).

Question 3

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Three.
Final answer: Three.

Question 4

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: [LT^-1].
Final answer: [LT^-1].

Question 5

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: [MLT^-2].
Final answer: [MLT^-2].

Question 6

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Yes.
Final answer: Yes.

Question 7

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: No.
Final answer: No.

Question 8

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: pascal (Pa).
Final answer: pascal (Pa).

Question 9

Working: Keep the first three significant digits: 1, 2, 4. The next digit is 8, so round 12.4 upward to 12.5.

Final answer: 12.5.

Question 10

Working: Momentum p=mv. Therefore [p]=M x LT^-1 = MLT^-1.

Final answer: [MLT^-1].

Question 11

Working: [s]=L. Also [ut]=LT^-1 x T=L, and [at^2]=LT^-2 x T^2=L. Every term has dimension L, so the equation is homogeneous.

Final answer: ut and at^2 both have dimension L; homogeneous.

Question 12

Working: 72 km/h x (1000 m/1 km) x (1 h/3600 s) = 72/3.6 = 20 m/s.

Final answer: 20 m s^-1.

Question 13

Working: In 5.070 x 10^3, the digits 5, 0, 7 and the final 0 are significant. Answer: 4 significant figures.

Final answer: Four.

Question 14

Working: Work = force x displacement; torque = force x perpendicular distance. Both have dimensions M L^2 T^-2, so dimensional analysis alone cannot distinguish their physical meanings.

Final answer: Both are [ML^2T^-2].

Question 15

Working: Assume T proportional to l^a g^b. Then T dimension = L^a(LT^-2)^b = L^(a+b)T^(-2b). Equate powers: a+b=0 and -2b=1. Hence b=-1/2 and a=1/2, so T proportional to sqrt(l/g).

Final answer: Equating dimensions gives a=, b=-, so T proportional to sqrt(l/g).

Question 16

Working: V=(4/3)pi r^3=(4/3)pi(2.00 cm)^3=33.510... cm^3. To 3 significant figures, V=33.5 cm^3.

Final answer: 4pir^3/3=33.5 cm^3.

Question 17

Working: From F=Gm1m2/r^2, G=Fr^2/(m1m2). Dimensions: (MLT^-2)L^2/M^2=M^-1L^3T^-2.

Final answer: [M^-1L^3T^-2].

Question 18

Working: Area=15.2 x 3.4=51.68 cm^2. The least precise factor has 2 significant figures, so report 52 cm^2.

Final answer: 51.68 cm^2 rounds to 52 cm^2.

Question 19

Working: Assume T proportional to m^a k^b. Since [k]=MT^-2, [T]=M^(a+b)T^(-2b). Set a+b=0 and -2b=1: b=-1/2, a=1/2. Thus T proportional to sqrt(m/k).

Final answer: a=, b=-; T proportional to sqrt(m/k).

Question 20

Working: Round 18.237 to 3 significant figures: retain 1, 8, 2; the next digit is 3, so the result is 18.2 cm^3.

Final answer: 18.2 cm^3; extra digits are unsupported.